

Four fundamental subspaces of a MatrixIndependence

- The columns of a matrix are independent if they are not the linear combination of other columns.

$$A_{m \times n}$$

$m < n$ implies there are more unknowns than eqⁿs

so there are non zero solutions to $Ax=0$ as there will be free variables and some columns will be combination of others.

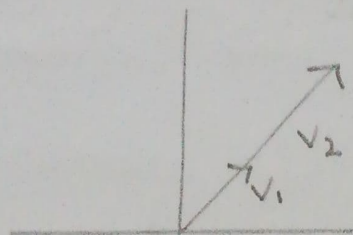
Formal Defⁿ

Vectors $x_1, x_2, \dots, x_n$ are independent if no combination

gives zero vector (except the zero combination)

→ If some other combination that zero gives a zero vector, then the vectors are dependent.

Example:



$$2v_1 = v_2$$

$$\therefore 2v_1 - v_2 = 0$$

$\therefore$ They are dependent vectors.

#

$$A = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \dots & \uparrow \\ v_1 & v_2 & v_3 & \dots & v_n \\ \downarrow & \downarrow & \downarrow & \dots & \downarrow \end{bmatrix}$$

matrix A or it's columns are independent if the nullspace of A is only the zero vector $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. (rank = $\text{col}^n(n)$)

Example:

$$\begin{bmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

No other condition will give the null space ; Independent

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 5 & 7 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Dependent column }

They are dependent if $Ac=0$ ($c \rightarrow$ column) for some non-zero c . ($\text{rank} < n$)

Independent $\rightarrow$ No column is the linear combination of any other column.

Dependent $\rightarrow$ If a column is the linear combination of other columns.

□ Span :

If the columns $v_1, v_2, v_3, \dots, v_n$ are independent, then the space that they cover is called the span of that matrix.

* If we have 3 vectors, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, the linear combination of these three vectors can cover the whole $\boxed{\mathbb{R}^3}$ space.

* For 3 vectors, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, in that case, we can't span $\mathbb{R}^3$, in that case the span will be a subspace (plane) in $\mathbb{R}^3$.

* An example of span is the columnspace of any matrix. This column space is spanned by the columns of that matrix.

Columnspace:

The span of the columns of a matrix

* The idea of basis comes from the idea of span.

The fundamental column vectors that spans any subspace or any vector space is known as the basis.

It is not possible to have 3 vectors in $\mathbb{R}^2$ that span the whole space. For example, the vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ do not span $\mathbb{R}^2$ because they are linearly dependent. In $\mathbb{R}^3$, 3 vectors can span the space. For example, the vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^3$.

Basis

The fundamental set of vectors that can span a space.

For example, for $\mathbb{R}^3$ the basis are $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

These are the fundamental set of vectors that can span $\mathbb{R}^3$.

* In $\mathbb{R}^n$ vectorspace, n vectors give basis, if the $n \times n$ matrix is invertible.

So, if we have an invertible (no. dependent col's) $n \times n$ matrix, it's columnspace will be $\mathbb{R}^n$.

In order to be a basis, for a space, we need 2 properties.

Basis for a space is a sequence of vectors $v_1, v_2, v_3, \dots, v_n$ with 2 properties.

1. The basis vectors have to be independent.

2. They have to span the whole space.

□ The basis of $\mathbb{R}^3$ are $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$,

Is the choice of picking basis vector for any space unique?

- No, as long as the two conditions are satisfied, the choice isn't unique.

For $\mathbb{R}^3$ we can choose any three vectors that can span the whole $\mathbb{R}^3$.

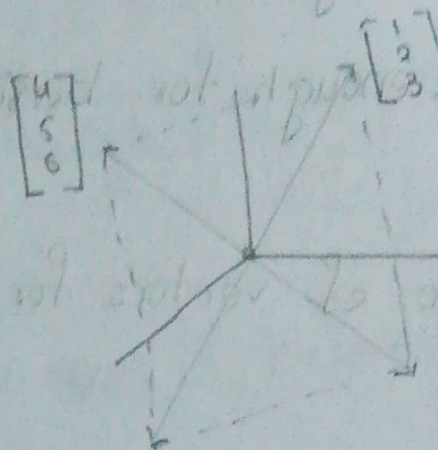
for example, $\mathbb{R}^3 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

as long as we can prove these 3 vectors are independent,

they can be the basis vector.

We need n vectors to span whole $\mathbb{R}^n$.

If we take two vectors, for example $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \right\}$
 will they still be the basis of $\mathbb{R}^3$?



The combination of these two vectors is a plane,
 to get whole $\mathbb{R}^3$ as span, we need another vector (at least)

So, no, though they are independent, they can't span $\mathbb{R}^3$.

But, they can be a basis for a subspace created by
 a plane in $\mathbb{R}^3$. (a subspace in $\mathbb{R}^3$)

Infinite combinations for a basis

"Every basis for a space has same number of vectors"

for R^6 ,

Taking 7 vectors is too many for the basis

Taking 5 vectors is not enough for basis

$\therefore$ 6 is the perfect no. of vectors for basis.

$\Rightarrow$ Dimension

Dimension:

The minimum number of vectors that can span the whole space,

And the set of vectors is called basis vector.

For R^n , n is the dimension.

$$\dim(C(A)) = \text{rank}$$

$$\dim(N(A)) = \# \text{ of free cols} = n - r$$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$\underbrace{\quad \quad \quad}_{c_1 + c_2}$
 c_1

* What is the span of these column vectors?

As there are two independent columns,

Generalised answer $\rightarrow$ the column space of A

Specific answer $\rightarrow$ The linear combination of $c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

$\rightarrow$ Dimension of Columnspace?

$$\dim(C(A)) / \text{independent cols} / \text{rank} = 2$$

* Are these columns independent?

No, 3 and 4 are obtained from 1 & 2 and 1 respectively

The null space of A has non-zero solutions as there are free columns here

* What is the basis of column space?

$$A \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$$

* What's the dimension of nullspace?

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

free columns

The two free columns gives us the whole nullspace
for these two columns we will get two special solutions

$$N(A) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Min no. of c's that gives us the nullspace = 2

$$\therefore \dim(N(A)) = 2$$

$$\dim(N(A)) = \# \text{ of free columns}$$

$$= n - r$$

The four fundamental subspaces of a matrix

1. Columnspace

2. Nullspace

3. Rowspace

4. Left Nullspace

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Example:

$$R = \begin{bmatrix} \underline{1} & 3 & 5 & 0 & 7 \\ 0 & 0 & 0 & \underline{1} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{\text{free}}$

$$\text{rank} = 2$$

$$\dim(C(A)) = 2$$

$$1. C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

2. Nullspace, $N(A)$

Basis $\rightarrow 3$ special

$$\left\{ \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{rank} = n - r = 5 - 2 = 3$$

$$\dim(N(A)) = 3$$

2. Row space $C(A^T)$

The space spanned by the rows of any matrix.

$$C(A^T) \quad [A^T \text{ rows} \Rightarrow \text{col's}]$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 0 \end{bmatrix}$$

$\checkmark \quad \checkmark \quad \times$

$$\text{rank} = 2 \quad (\# \text{ of pivot col's})$$

$$\dim(C(A^T)) = 2$$

$$\text{Basis} = \left[\begin{bmatrix} 1 \\ 3 \\ 5 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right]$$

columnspace of A^T
 $\rightarrow$ row space

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

4. Left nullspace $N(A^T)$

The nullspace of row space

$$Ay = 0$$

$y \rightarrow$ Nullspace / special 31

$$\Rightarrow (Ay)^T = 0$$

$$\Rightarrow y^T A^T = 0$$

row space
 Nullspace comes left.

$$R^T = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 0 \end{bmatrix}$$

Pivot
Free

$$N(R^T) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\dim(N(A^T)) = \# \text{ of free cd}^n \text{ in } A^T = m-r$$

$$= 1$$

If A is $m \times n$ matrix with rank r , its left subspace has dimension $\boxed{m-r}$

[Here $m \times n$ is of A , not A^T]

Independent row $\rightarrow$ non zero value

Dependent row $\rightarrow$ get zeros (for row operation)

$$\# \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ x & x & x & x \end{bmatrix} \begin{bmatrix} x \\ x \\ x \\ x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

If we have a $m \times n$ matrix,
there are 3 elements in each column

- $\therefore$ The $C(A)$ would be in $\mathbb{R}^m$
- The $C(A^T)$ would be in $\mathbb{R}^n$.
- The $N(A)$ would be in $\mathbb{R}^n$
- The $N(A^T)$ would be in $\mathbb{R}^m$

In $\mathbb{R}^n$, $\dim(C(A^T)) = r$ [row space]
 $\dim(N(A)) = n - r$ [nullspace]

In $\mathbb{R}^m$, $\dim(C(A)) = r$ [colⁿ space]
 $\dim(N(A^T)) = m - r$ [left nullspace]

[same vector space]

Columnspace

$C(A)$

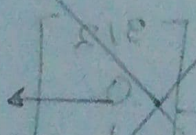
$\dim = r$

Rowspace

$C(A^T)$

$\dim = r$

R^m



Big x

PICTURE

Left Nullspace

$N(A^T)$

$\dim = m - r$

Null space

$N(A)$

$\dim = n - r$

for $m \times n$ matrix
of rank r

$$\# \text{Dim}(C(A)) = \text{Dim}(C(A^T)) = r$$

$$\# \text{Dim}(N(A)) = \text{Dim}(N(A^T)) = m - r$$

$$r + (m - r) = m$$

Example:

$$A = \begin{bmatrix} 2 & -1 & -3 \\ -4 & 2 & 6 \end{bmatrix}$$

$\begin{matrix} \checkmark & & \times \\ r = \text{mib} & & \times \end{matrix}$

$$A^T = \begin{bmatrix} 2 & -4 \\ -1 & 2 \\ -3 & 6 \end{bmatrix}$$

$\begin{matrix} \checkmark & & \times \\ r = \text{mib} & & \times \end{matrix}$

$r = 1, m = 2, n = 3$

Columnspace

$$\text{span} = \left\{ \begin{bmatrix} 2 \\ -4 \end{bmatrix} \right\}$$

$$\text{basis} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

$$\dim = 1$$

Nullspace

$$x_s = \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3/2 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Span} = \left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3/2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\dim = 2$$

Rowspace

$$\text{span} = \left\{ \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix} \right\}$$

basis

$$\dim = 1$$

Left Nullspace

$$x_s = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{r.r.e.} = \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{span} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$$

$$\dim = 1$$

$$\# \dim(C(A)) = \dim(C(A^T)) = r$$

$$\# \dim(N(A)) = n - r \quad \left| \quad \dim(N(A^T)) = m - r$$